

Design, Fabrication and Analysis of three finger gripper actuated by micro servos

A project report

Submitted

In Partial Fulfillment of the Requirement of the Degree

Of

Bachelor of Technology



By

Souvik Poddar (14UPE045)

MD. Afzal Hussain (14UPE058)

Supratim Sarkar(14UPE021)

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Under the supervision of

Dr. Kishan Choudhuri

DEPARTMENT OF PRODUCTION ENGINEERING

NATIONAL INSTITUTE OF TECHNOLOGY AGARTALA

DECEMBER 2017

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Approval Sheet

This project report titled “**Design, Fabrication and Analysis of three finger gripper actuated by micro servos**” by Biltu Sarkar (14UPE030), Souvik Poddar (14UPE0), Supratim Sarkar (14UPE021), Afzal Hussain (14UPE0), Rohit Kumar (14UPE063) is approved for the degree of B.Tech, Production Engineering.

Examiners

Supervisor

Date:

Place:

Declaration

I declare that this written submission represents my ideas in my own words and where others ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misinterpreted or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Souvik Poddar(14UPE045)

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Place:

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B.TECH (VIII Semester)

Department of Prouction Engineering
National Institute of Technology Agartala

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Certificate

This is to certify that this is a bonafide record of the project titled “Design, Fabrication and Analysis of three finger gripper actuated by micro servos” presented by Souvik Poddar (14UPE045), MD Afzal Hussain (14UPE058), Supratim Sarkar (14UPE021), Biltu Sarkar (14UPE030), Rohit Kumar (14UPE063) in partial fulfilment of the requirements of the degree of Bachelor of Technology in Production Engineering. He has fulfilled the requirements for the submission of this project, which to the best of my knowledge has reached the required standard. This project is carried out under my supervision and guidance and has not been submitted elsewhere for the award of any other degree.

Date:

Dr.Kishan Choudhuri
(Project Guide)

Acknowledgement

I feel immense pleasure and privilege to express my deep sense of gratitude, indebtedness and thankfulness towards my superior Dr. Kishan Choudhuri, Department of Production Engineering, National Institute of Technology Agartala, for his invaluable guidance, constant supervision, continuous encouragement and support throughout this work. His suggestions and critical views have greatly helped me in different steps of literature review, problem understanding and its methodology for implementation in the project. I am also thankful to Mr. Prasenjit Debnath, Department of Production Engineering, and other management personnel for supporting me with all required infrastructural facilities required for implementing the project work.

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Abstract

Research works in the field of grippers have been carried out extensively at various countries. Focus has been put extensively on the application of various materials for fabrication of grippers and fingers as well as the type of actuation mechanism that the finger or gripper is going to use for its operation. Focus has also been laid on finding the forward and reverse kinematics model of the grippers. Research works are primarily based on fingers and not the grippers as a whole. Keeping this mind we have decided to focus on the gripper as whole as not a lot of research works have been carried out extensively on this field. Also research works have stated that three finger gripper models are far more useful and beneficial for use than two finger or five finger grippers that have mostly been manufactured. Based on the above ideas we have fabricated a three finger gripper with aluminium as the finger material and servo motor being the actuators. We have given three links and two joints on each of the finger each joint being actuated by servo mechanism. The whole assembly has been assembled and put on a tripod stand for carrying out further analysis of the model.

Contents

	Page No.
<i>Approval Sheet</i>	3
<i>Declaration</i>	4
<i>Certificate</i>	5
<i>Acknowledgement</i>	6
<i>Abstract</i>	7
1. Literary articles and objectives of present study	
1.1 Literary articles	9-11
1.2 Objectives	12
2. Introduction	
2.1 Robotic Grippers	13-17
2.2 Arduino	18-22
2.3 Servo motors	22-24
2.4 LabVIEW	24
2.5 NI ELVIS	25
3. CAD Designing	26-30
4. Required components and experimental setup	
4.1 Electronics and LabVIEW setup	31-33
4.2 Mechanical Model	33-36
4.3 Programming	36-38
5. Experimental Results	39-43
6. Theoretical Explanation	44-46
7. References	47

CHAPTER 1

Literary Articles and Objective of the present study

1.1 Literary Articles

Some of the research articles are discussed below which worked as motivation of the present study.

Jian Huang, Bui Trong Quan, Minoru Harada, Tetsuro Yabuta

Emulating the Motion of a Human Upper Limb: Controlling a Finger-arm Robot by using the Manipulability of its Finger

In this paper, a 3-DOF robot finger is fixed onto the end-effector of a 6-DOF robot arm to realize a finger-arm robot. As in the case of the upper limb of a human being, the finger-arm robot possesses a high degree of freedom. This paper describes a new method to control the finger-arm robot so as to simply generate a natural motion emulating the movement of a human upper limb. To achieve this, a control method based on the manipulability of the finger is proposed. By using the proposed method, the finger primarily moves to complete a delicate local task, while the arm only moves to assist the finger when the manipulability of the finger falls below a reference value. The proposed method was applied to the task of drawing two 3D figures. The obtained results clarify the relation between manipulability and kinetic energy and also demonstrate the effectiveness of the proposed method.

Young June Shin, Ho ju Lee, Kyun- Soo Kim, Soohyun Kim

A Robot Finger Design Using a Dual-Mode Twisting Mechanism to Achieve High-Speed Motion and Large Grasping Force

A dual-mode robot finger is proposed to achieve a high-speed motion and large grasping force with a single motor. The robot finger has two actuator modes, which consist of the speed mode and the force mode. Based on the geometric analysis of each mode, the main design parameters of the proposed robot finger are derived, and their effectiveness is verified by simulations. In addition, using experiments with a prototype of a robot finger, the validity of the proposed approach is demonstrated.

Ikuo Yano, Kenjiro Takemura and Takashi Maeno

Development of a Robot Finger for Five-fingered Hand using Ultrasonic Motors

A robot finger is developed for five-fingered robotic hand and having equal number of DOF to human hand. The robot hand is driven by a new method proposed by authors using ultrasonic motors and elastic element. The method utilizes restoring force of elastic element as driving power for grasping an object, so that the hand can perform the soft and stable grasping motion with no power supply. In addition, all the components are placed inside the hand thanks to the ultrasonic motors with compact size and high torque at low speed. Applying the driving method to multi-DOF mechanism, a robot index finger is designed and implemented. It has equal number of joints and DOF to human index finger, and it is also equal in size to the finger of average adult male. The performance of the robot finger is confirmed by fundamental driving test.

Qi Zhou, Hongyi Li, and Peng Shi

Decentralized Adaptive Fuzzy Tracking Control for Robot Finger Dynamics

In this paper, a novel design method for adaptive tracking control is proposed for robot finger dynamics. First, the dynamics are described by considering the robot finger as a largescale system since it has many joints and multi-degrees of freedoms (DOFs). Second, by employing the direct adaptive fuzzy approximation method to approximate the unknown and desired control input signals instead of the unknown nonlinear functions, the number of adaptive design parameters obtained in the control design process is greatly reduced. Moreover, it is shown that the designed controller can guarantee all the signals in the closed-loop system to be semiglobally uniformly ultimately bounded. Finally, simulations are conducted on the Puma 560 robot manipulator, and the results show the effectiveness of the developed control design approach.

Li Jiang, Dong Sun

An Inverse-Kinematics Table-Based Solution of a Humanoid Robot Finger With Nonlinearly Coupled Joints

This paper presents a novel approach toward the inverse kinematics solution of the humanoid robot fingers with nonlinearly coupled joints, a challenging problem existing for years. Under an assumption that the coupled joint angles of the finger are the same, we first derive an approximate closed-form solution of the finger's inverse kinematics. Then, utilizing the approximate solution as the ancillary variable, we propose to solve the finger joint angles from this approximate solution rather than the fingertip position. Through analyzing properties of the approximate solution, it can be known that the coupled joint angles in the approximate solution play the most important role in the derivation of inverse kinematics. In practical implementation, a 1-D lookup table and the linear interpolation to the approximate solution are used to calculate the accurate joint angles. Simulation and experimental results demonstrate effectiveness of the proposed inverse kinematics method.

Shinya Kajikawa

Robot Finger Module With Multidirectional Adjustable Joint Stiffness

This paper describes a novel finger mechanism that enables a robot to perform a gentle and dexterous motion. This finger contains three joints, in which the proximal (PIP) and distal intraphalangeal (DIP) joints are coupled rigidly, and the metacarpophalangeal (MP) joint has a multidirectional passive compliance that is adjustable as in humans. In the MP joint, two kinds of silicone-rubber cushions (A-SRC and B-SRC) are placed between a motor-driven disk and an output link. A-SRC plays three roles 1) transmission of the rotational motion from the motor to the output link by the friction force between the surface of A-SRC and the motor-driven disk; 2) absorption of external forces from multiple directions by its elastic deformation; and 3) sensing the directions and magnitudes of external forces with the change in inside pressure. B-SRC adjusts the characteristics of joint compliance by pressing A-SRC against the motor-driven disk. We investigate the basic performance of this finger module through several experiments.

Srijan Bhattacharya, Bikash Bepari, Subhasis Bhaumik

Soft Robotic Finger Fabrication with PDMS and IPMC Actuator for Gripping

Scientific community has become very contemplative about the new breakthrough of micro scale actuations in engineering and biomedical applications, such micro systems with micro actuators are becoming very useful for handling and manipulating micro objects including biological cells. For the development of such devices, it is important to select proper materials, actuators and sensing elements which play an important role in designing micro gripper. In the present

discussion a survey of micro gripping is highlighted. At the same time a unique approach of micro finger design with Polydimethyl Siloxane (PDMS) is reported with experimental results. Those results are shown while the Ionic Polymer Metal Composite (IPMC) actuators are excited with mechanical and electrical stimuli.

Z. Doulgeri_, Y. Karayiannidis

Force position control for a robot finger with a soft tip and kinematic uncertainties

We consider the problem of force and position regulation for a robot finger with a soft tip in contact with a surface with unknown geometrical characteristics. An adaptive controller is proposed, and the asymptotic convergence of the applied force error and the estimated position error of the tip to zero is shown for the spatial case. Simulation results demonstrate the controller performance.

Haruhisa Kawasaki, Tsuneo Komatsu, and Kazunao Uchiyama

Dexterous Anthropomorphic Robot Hand With Distributed Tactile Sensor: Gifu Hand II

This paper presents an anthropomorphic robot hand called the Gifu hand II, which has a thumb and four fingers, all the joints of which are driven by servomotors built into the fingers and the palm. The thumb has four joints with four-degrees-of-freedom (DOF); the other fingers have four joints with 3-DOF; and two axes of the joints near the palm cross orthogonally at one point, as is the case in the human hand. The Gifu hand II can be equipped with six-axes force sensor at each fingertip and a developed distributed tactile sensor with 624 detecting points on its surface. The design concepts and the specifications of the Gifu hand II, the basic characteristics

of the tactile sensor, and the pressure distributions at the time of object grasping are described and discussed herein. Our results demonstrate that the Gifu hand II has a high potential to perform dexterous object manipulations like the human hand.

Akitoshi Harada, Takashi Nakakuki, Mitsutaka Hikita and Chiharu Ishii

Robot Finger Design for Myoelectric Prosthetic Hand and Recognition of Finger Motions via Surface EMG

In this paper, robot hand design aiming at application to a myoelectric prosthetic hand and recognition of finger operation via surface EMG are discussed. The robot hand with two fingers, thumb and index finger, was built to implement fundamental motions required in daily life, such as grasping and holding. The fingers are driven by the wire actuation system to imitate the human's tendon mechanism. In order to control each finger of the developed myoelectric prosthetic hand independently, using the neural network, identifier which classifies four finger operations, namely flexion and extension of the thumb in metacarpophalangeal (MP) joint and flexion and extension of the index finger in MP joint, was constructed. The recognition of each finger operation was performed using the four patterns of the neural network based identifiers in which the recognition method of finger operations differs each other, and the results were compared.

1.2 OBJECTIVES

The objective of the current project is to study the servo motor characteristics when it is subjected to a load gripped by the gripper. We want to see how the voltage fluctuates with respect to time and also determine the maximum load that the gripper can grip as a whole.

The whole model of the gripper has been analysed based on how much load it can carry and the variation of the input voltage of the servo motor with respect to load.

For studying the above characteristics, a setup was prepared which will be used for carrying the load. The setup prepared is shown below. It was prepared keeping in mind that it's own weight should not be significant compared with the applied weight.

1. MAXIMUM LOAD

The load setup was attached to the gripper which was in locked position. The load was gradually increased until the setup starts to slip. It was found out that when the applied load was increased to 50 kg, the setup started to slip. Hence, with this method we calculated the load bearing capacity of the gripper which came out to be 50 kg.

2. VOLTAGE VERSUS LOAD CHARACTERISTICS WITH TIME

A sample of 125 data has been collected for each of the applied load which in value is a multiple of five starting from 5 to 40. One of the servo motor of the lower joint has been connected to ELVIS and the voltage versus load characteristics has been studied using LABVIEW software. The sample data was stored in an excel file and a graph was plotted having voltage on the Y- axis and load on the X- axis.

CHAPTER 2:

Introduction

2.1 Robotic grippers

Grippers, sometimes called hand grippers, are primarily used for testing and increasing the strength of the hands; this specific form of grip strength has been called crushing grip,^[1] which has been defined as meaning the prime movers are the four fingers, rather than the thumb.^[2]

There are differences from brand to brand, but the common features of standard grippers are that they use a torsion spring fitted with two handles. The exact dimensions of these elements vary, as well as the materials used to make them; the springs are made from various types of steel, and the handles are generally made from wood, plastic, steel or aluminum.

The force that a robotic gripper applies to a part is typically used by engineers to select grippers. While gripper force is a first order consideration, the torque that is experienced by the gripper is equally as critical and, unfortunately, usually only addressed in a cursory manner. In some cases the torque is addressed via the gripper manufacturer supplying jaw length vs. force charts. These gripper charts are helpful but are only useful in low G-force applications and provide rough guidance at best. The result of this situation is that “rookie” gripper application engineers end up with dropped parts and “old hands” end up with grippers far larger and more expensive than required.

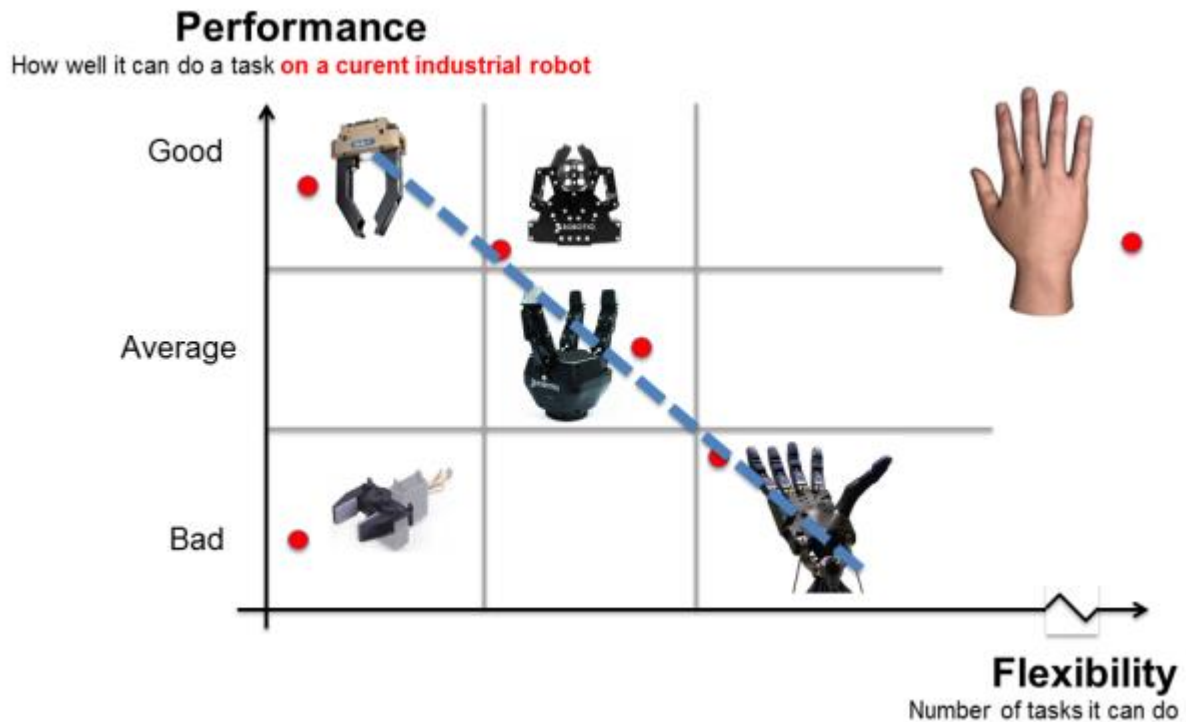
Gripper Slide Efficiency

The efficiency of a gripper's slide is often provided via a chart that shows jaw length versus force provided. This is an indication of the amount of force actually provided by the gripper versus the theoretical force provided by the pneumatic or electric system. While efficiency is interesting, in most cases it has little to do with the size requirement of a gripper because the slide is equally efficient in both directions. Therefore, if the part is pushing back on the jaw, due to weight or acceleration, the slide will be as efficient or inefficient transmitting this part-created force to the gripper as it is in transmitting the gripper-created force to the part.

Slide efficiency really only comes into play when the gripper jaws are designed so that only a friction grip is provided and “squeeze” is the only thing holding in the part (jaw types and their effect on size requirements are provided in detail later). Typically the slide efficiency charts are only a rough measure of the slide's torque capability. Therefore, while the charts may be helpful, the better gripper manufacturers now provide torque specifications for pitch jaw and roll of the jaws. Bottom line...ask for torque information when sizing a gripper.

Why three finger grippers are used?

Well to analyze this scientifically, we have to evaluate the other options too, i.e. other than using a 3 fingered gripper. So let's see what is available on the market.



One Finger Grippers

Basically a hook. This is probably the simplest way to put a gripper on your robot. Even if you think it sounds ridiculous, some applications simply need a hook or probe to achieve what they need to do. In terms of precision and repeatability it is probably not very accurate, but it is also the cheapest you can get, no power needed, no maintenance. Look at the following example that uses a hook to bring a part out of the CNC machine in a machine tending operation.

Flexibility: +

Repeatability: +

Price: +

Two Finger Grippers

Parallel



There are different kinds of two fingered grippers on the market. The most popular ones are the parallel grippers. These grippers are either electrical, pneumatic or hydraulic and use two jaws that close against each other and remain parallel to each other. This means that only a pinch grasp can be accomplished using these grippers. These grippers are then really limited and are really rigid. In fact, they can rarely be programmed to adapt to the shape of the object. So custom fingertips and/or custom jaws have to be made when a new production or a different product has to be handled with the gripper.

The advantage of having 2 fingers in this scenario is that you can roughly locate the part in the gripper. You can assume that the part will always be between the fingers, so it is relatively easy to automate a process knowing that.

Flexibility: +

Repeatability: ++++

Price*: +++++

* Pricing is mostly related to tooling.

Adaptive



These grippers are a little more exotic than the standard parallel grippers. They can be used in a production that requires more flexibility. In fact, these grippers can adapt themselves to the shape of the object it is grasping (rectangular, cylinder, etc.). In other words, if the production first involves a small rectangular part and then later on a round part, there is no need to change the gripper, it can adapt itself in the blink of an eye.

Having a 2-Finger Adaptive Gripper means an automatic repositioning of the part in the Gripper. The Gripper will be in parallel grasping mode or encompassing mode, either way the part will be where it should be. In terms of robot programming this makes things easier since your part is always located at the same place in the Gripper.

Three Finger Grippers

Parallel



These grippers are pretty much similar to a turning machine chuck fitted on a robot, they are also known as self-centering. The 3 "fingers" are closing toward a central axis to squeeze the part at the very center of this axis. Most of these grippers are actuated by hydraulic or pneumatic energy. They can usually carry a large payload, but are not very flexible at all and you have to change fingers or jaws for each different application.

Flexibility: ++

Repeatability: +++++

Price: +++

Adaptive



Using a 3-Finger Adaptive Gripper means that you will have a greater flexibility and more reliability to grip or grasp different kinds of parts. In fact, with under actuated fingers, the Gripper can adapt its configuration to the part geometry. The Gripper is also able to carry a relatively large payload for its weight (structural to payload ratio). This Gripper has 3 different grasping modes (scissor, wide and regular) which gives you a wide range of grasping setups for all kinds of applications. Just to put another spin on it, this Gripper has been used by 8 teams at the DARPA Robotics Challenge to achieve manipulation tasks for disaster relief and it was also able to absorb huge impacts from the 200 kg robots when they fell on the Gripper.

The advantage of such a Gripper is the level of flexibility that can be accessible for any given application. Whether it is for research or industrial applications, it can adapt to pretty much everything. It can also handle a payload of 10 kg. The positioning in the Gripper highly depends

on the grasping mode you have chosen, but is generally pretty accurate and has a repeatability of less than 0.05 mm.

Flexibility: +++++

Repeatability: +++++

Price: +++++

Four-five Finger Grippers



These type of grippers are still pretty unusual for any type of application. Even if they have been proven to work as a prosthetic for the human body, it is quite tough to make this work on a robot. So these grippers, most of the time, are very flexible because they can grasp a LOT of stuff, but they do not often have an accurate repeatability and they often cannot handle a heavy payload. This why these kinds of grippers are not very popular, they are just not ready yet for industry.

Flexibility: +++++

Repeatability: ++

Price: +++++

So if we are making a short resume of all kinds of 'fingered' grippers. Single fingered grippers are cheap but are not too precise, they can work well for removing a part from a machine for example. Two finger grippers are better because they can grasp a wider range of parts and handle them with better precision and their payloads are descent, but it can be complicated to use a parallel gripper when changing part size or geometry. The three fingered grippers are more precise and can handle more diverse part sizes and shapes. Thus, they are generally more expensive. Grippers with more than 4 fingers are complicated to use and cannot carry a heavy payload, so they are still in the research stage and are not really used in industry.

2.2 Arduino

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

Over the years Arduino has been the brain of thousands of projects, from everyday objects to complex scientific instruments. A worldwide community of makers - students, hobbyists, artists, programmers, and professionals - has gathered around this open-source platform, their contributions have added up to an incredible amount of accessible knowledge that can be of great help to novices and experts alike.

What you will need:

- A computer (Windows, Mac, or Linux)
- An Arduino-compatible microcontroller (anything from this guide should work)
- A USB A-to-B cable, or another appropriate way to connect your Arduino-compatible microcontroller to your computer (check out this [USB buying guide](#) if you're not sure which cable to get).



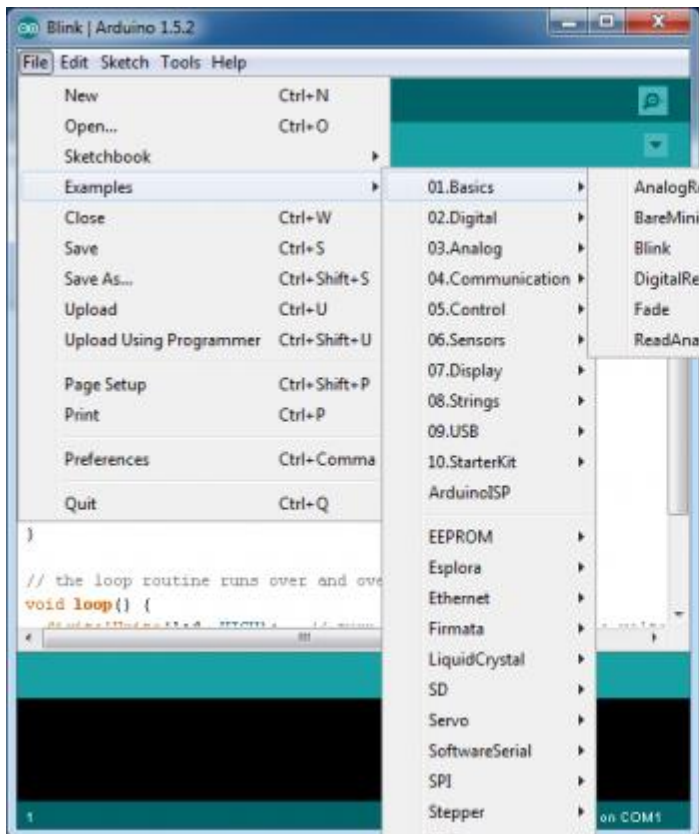
Arduino UNO



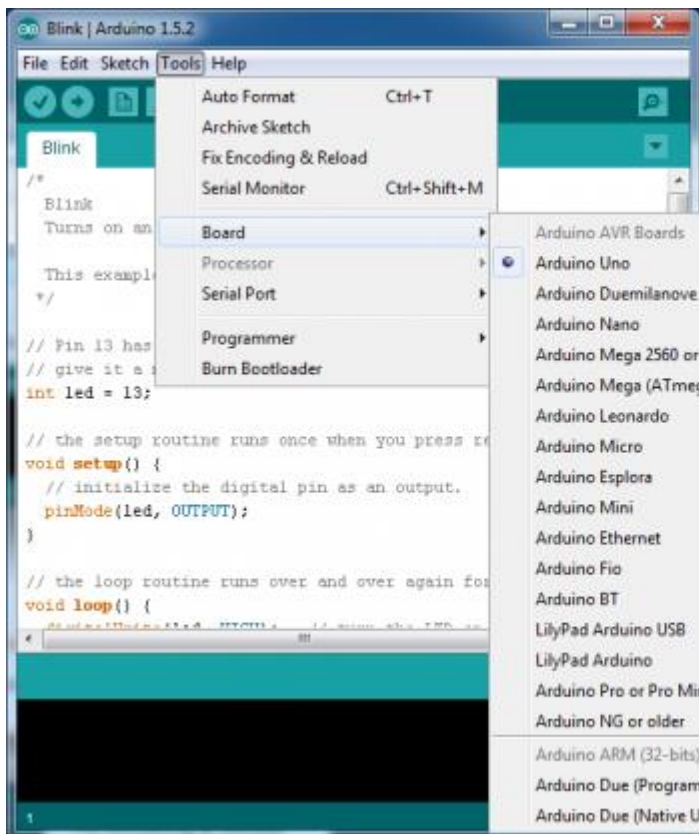
USB cable

Procedure of programming arduino

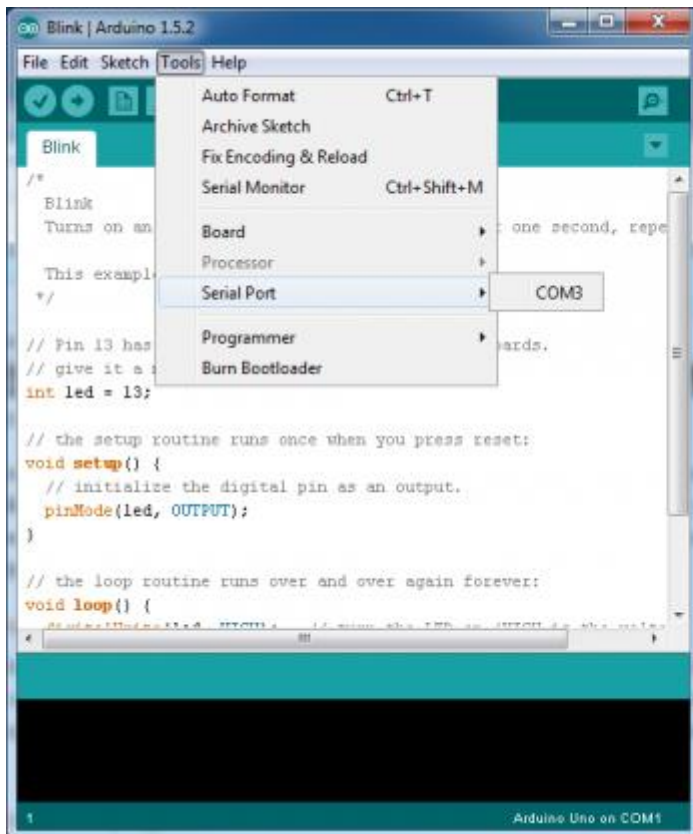
- Launch the Arduino application
- If you disconnected your board, plug it back in
- Open the Blink example sketch by going to: File > Examples > 1.Basics > Blink



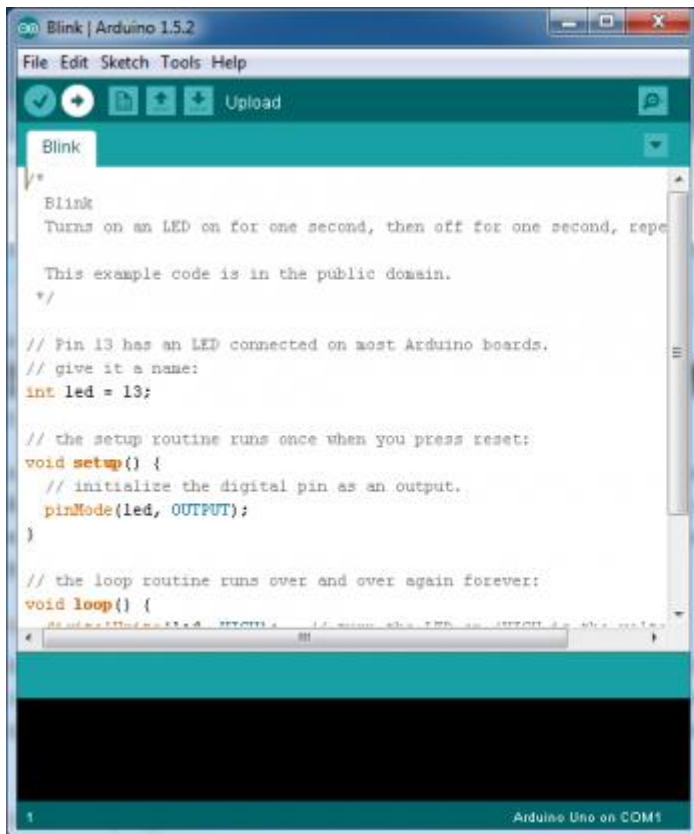
- Select the type of Arduino board you're using: Tools > Board > your board type



- Select the serial/COM port that your Arduino is attached to: Tools > Port > COMxx



- If you're not sure which serial device is your Arduino, take a look at the available ports, then unplug your Arduino and look again. The one that disappeared is your Arduino.
- With your Arduino board connected, and the Blink sketch open, press the 'Upload' button



- After a second, you should see some LEDs flashing on your Arduino, followed by the message ‘Done Uploading’ in the status bar of the Blink sketch.

2.3 Servo motors

A servo motor is one of the widely used variable speed drives in industrial production and process automation and building technology worldwide. Although servo motors are not a specific class of motor, they are intended and designed to use in motion control applications which require high accuracy positioning, quick reversing and exceptional performance.

What are Servo Motors?

A servo motor is a linear or rotary actuator that provides fast precision position control for closed-loop position control applications. Unlike large industrial motors, a servo motor is not used for continuous energy conversion.

Servo motors have a high speed response due to low inertia and are designed with small diameter and long rotor length. Then how do servo motors work?

Servo motors work on servo mechanism that uses position feedback to control the speed and final position of the motor. Internally, a servo motor combines a motor, feedback circuit, controller and other electronic circuit.

It uses encoder or speed sensor to provide speed feedback and position. This feedback signal is compared with input command position (desired position of the motor corresponding to a load), and produces the error signal (if there exist a difference between them).

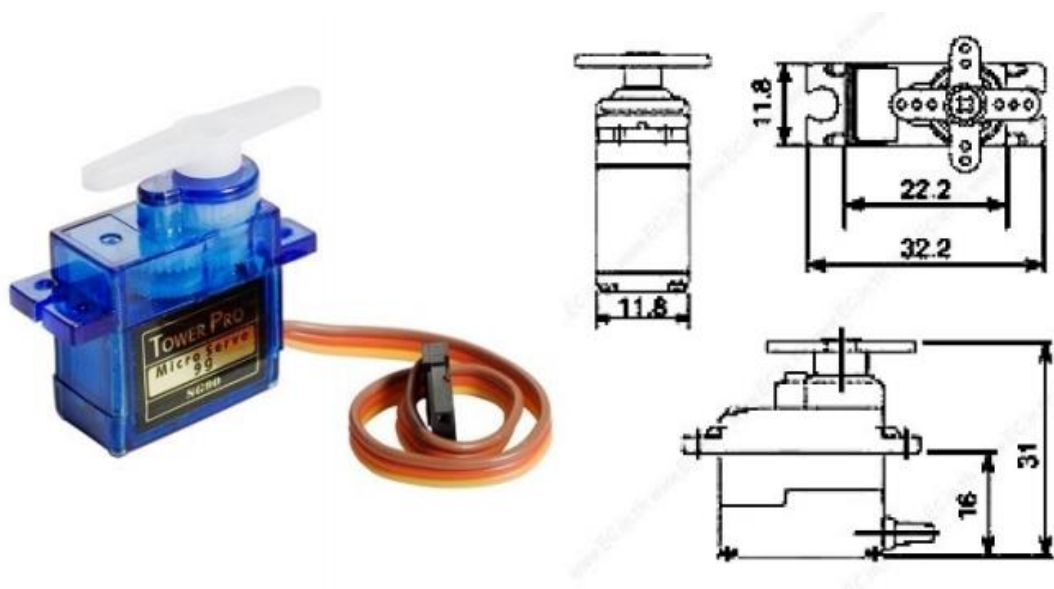
The error signal available at the output of error detector is not enough to drive the motor. So the error detector followed by a servo amplifier raises the voltage and power level of the error signal and then turns the shaft of the motor to desired position.

Types of Servo Motors

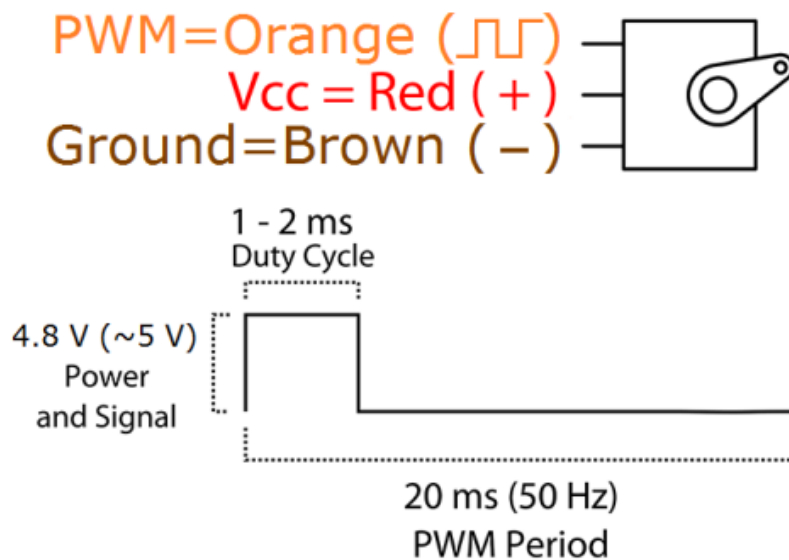
Basically, servo motors are classified into AC and DC servo motors depending upon the nature of supply used for its operation. Brushed permanent magnet DC servo motors are used for simple applications owing to their cost, efficiency and simplicity.

These are best suited for smaller applications. With the advancement of microprocessor and power transistor, AC servo motors are used more often due to their high accuracy control.

SG90 9 g Micro Servo



Tiny and lightweight with high output power. Servo can rotate approximately 180 degrees (90 in each direction), and works just like the standard kinds but smaller. You can use any servo code, hardware or library to control these servos. Good for beginners who want to make stuff move without building a motor controller with feedback & gear box, especially since it will fit in small places. It comes with a 3 horns (arms) and hardware.



Position "0" (1.5 ms pulse) is middle, "90" (~2 ms pulse) is all the way to the left. ms pulse) is middle, "90" (~2 ms pulse) is all the way to the right, " the

Specifications

- Weight: 9 g
- Dimension: 22.2 x 11.8 x 31 mm approx.
- Stall torque: 1.8 kgf·cm
- Operating speed: 0.1 s/60 degree
- Operating voltage: 4.8 V (~5V)
- Dead band width: 10 μ s
- Temperature range: 0 °C – 55 °C

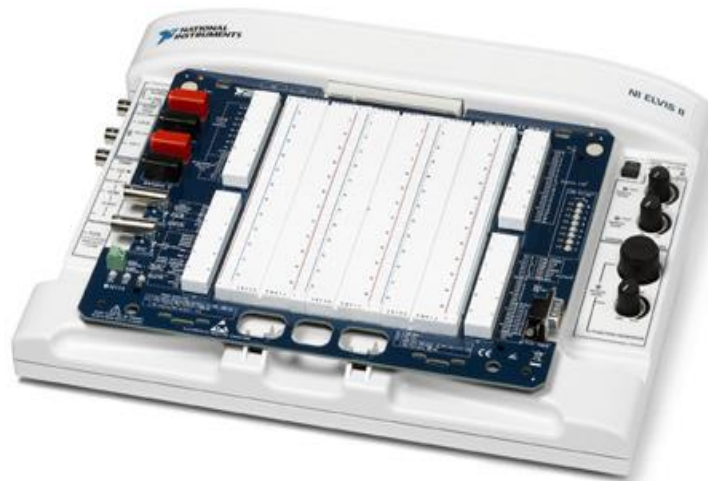
2.4 LabVIEW



LabVIEW, short for *Laboratory Virtual Instrument Engineering Workbench*, is a programming environment in which you create programs using a graphical notation (connecting functional nodes via wires through which data flows); in this regard, it differs from traditional programming languages like C, C++, or Java, in which you program with text. However, LabVIEW is much more than a programming language. It is an interactive program development and execution

system designed for people, like scientists and engineers, who need to program as part of their jobs. The LabVIEW development environment works on computers running Windows, Mac OS X, or Linux. LabVIEW can create programs that run on those platforms, as well as Microsoft Pocket PC, Microsoft Windows CE, Palm OS, and a variety of embedded platforms, including Field Programmable Gate Arrays (FPGAs), Digital Signal Processors (DSPs), and microprocessors.

2.5 NI ELVIS



The NI Educational Laboratory Virtual Instrumentation Suite (NI ELVIS) is a modular engineering educational laboratory device developed specifically for academia. With its hands-on approach, educators can help students learn practical, experimental engineering skills. The NI ELVIS includes an oscilloscope, digital multimeter, function generator, variable power supply, Bode analyzer, and other common lab instruments. You can connect a PC to the NI ELVIS using USB and build circuits on its detachable protoboard.

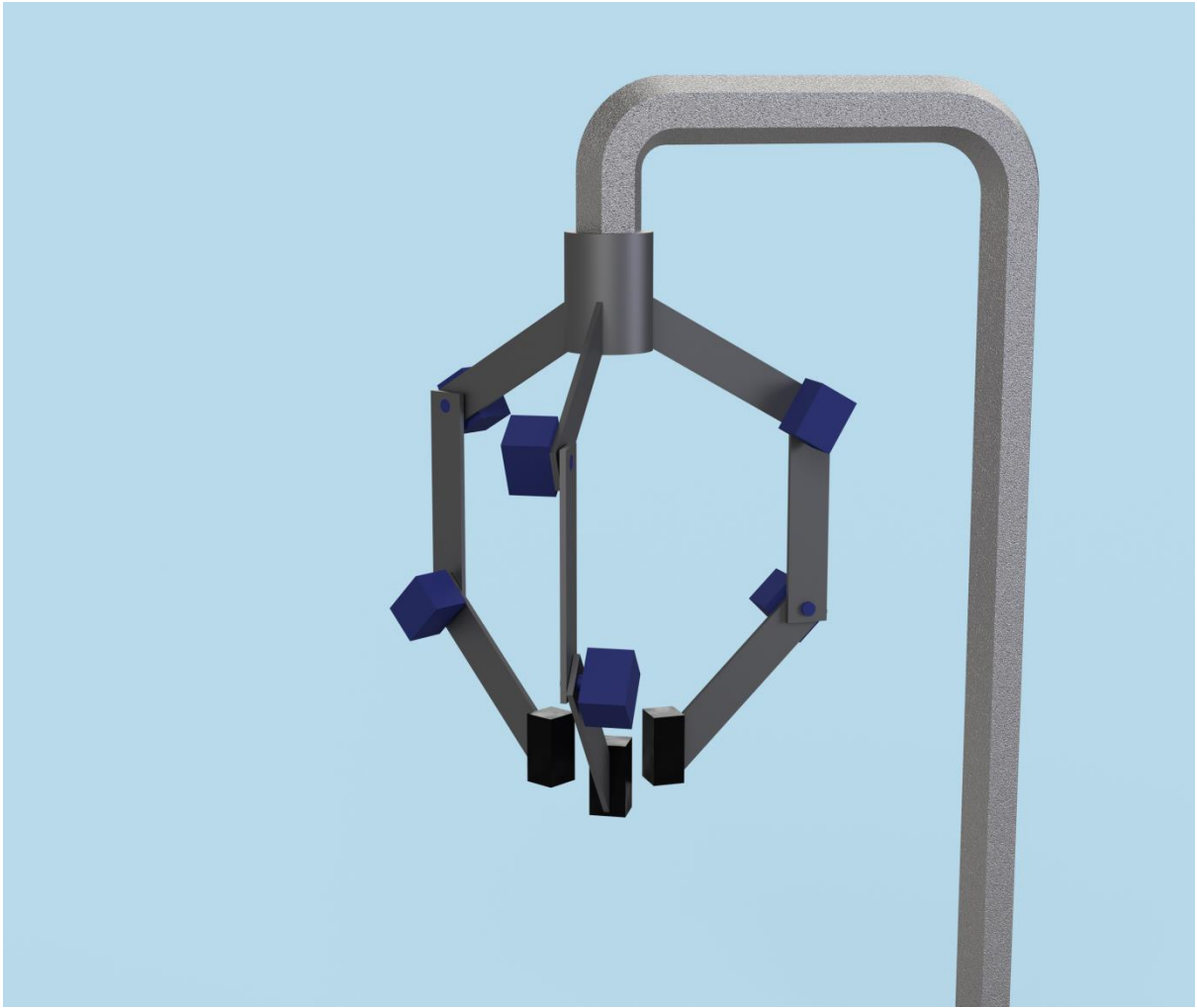
CHAPTER 3:

CAD Designing

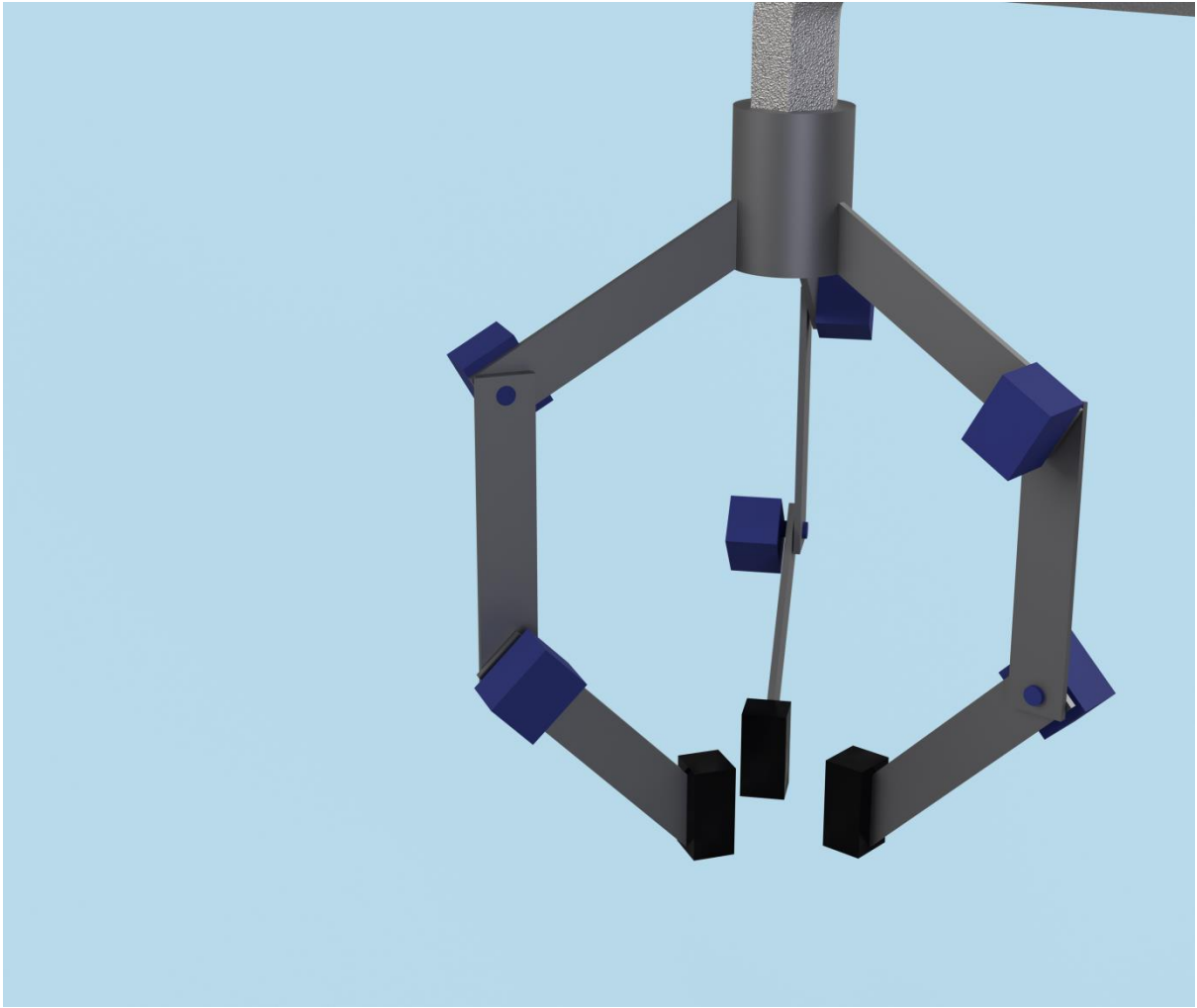
Before fabricating the actual model of the gripper, we have made a CAD model to get an idea of how the actual gripper would look like. The CAD model was designed on CATIA V5. The blue blocks on the joint of the gripper are the micro servos and the black blocks at the end effectors are rubber pads. We have tried to approximate the CAD model as much as possible in reality. The real life picture of the gripper is also shown below.



FINAL ASSEMBLY



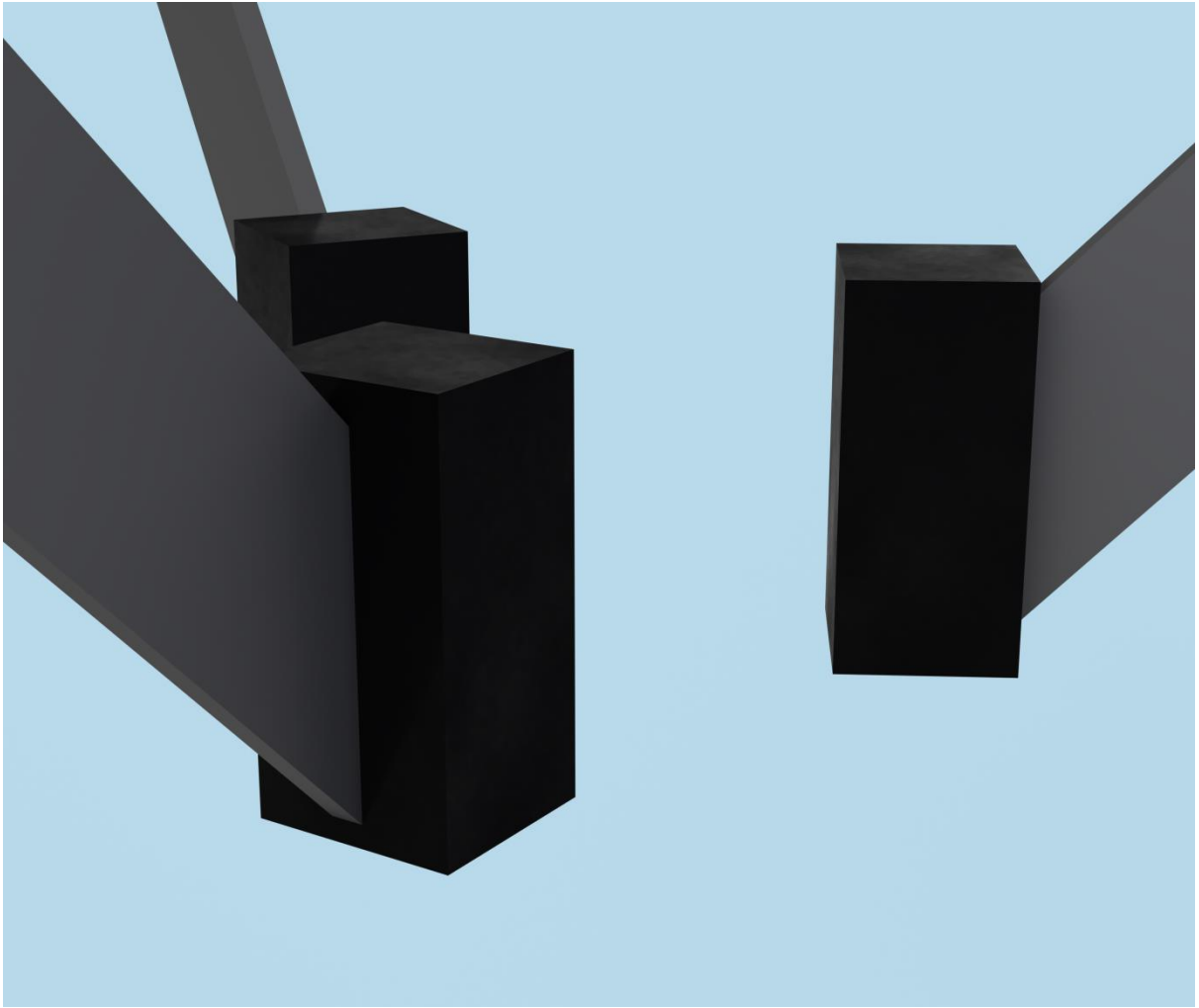
SIDE VIEW



SIDE VIEW ZOOMED



BOTTOM UP VIEW



END EFFECTOR

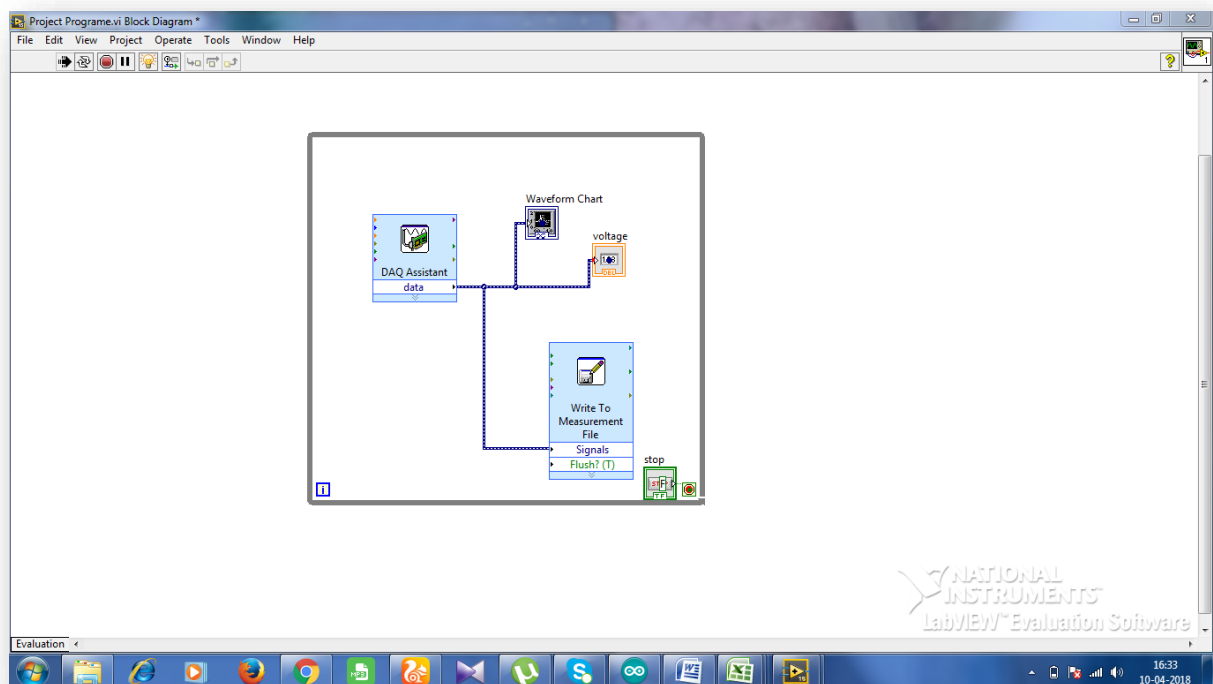
CHAPTER 4:

Required components and experimental setup

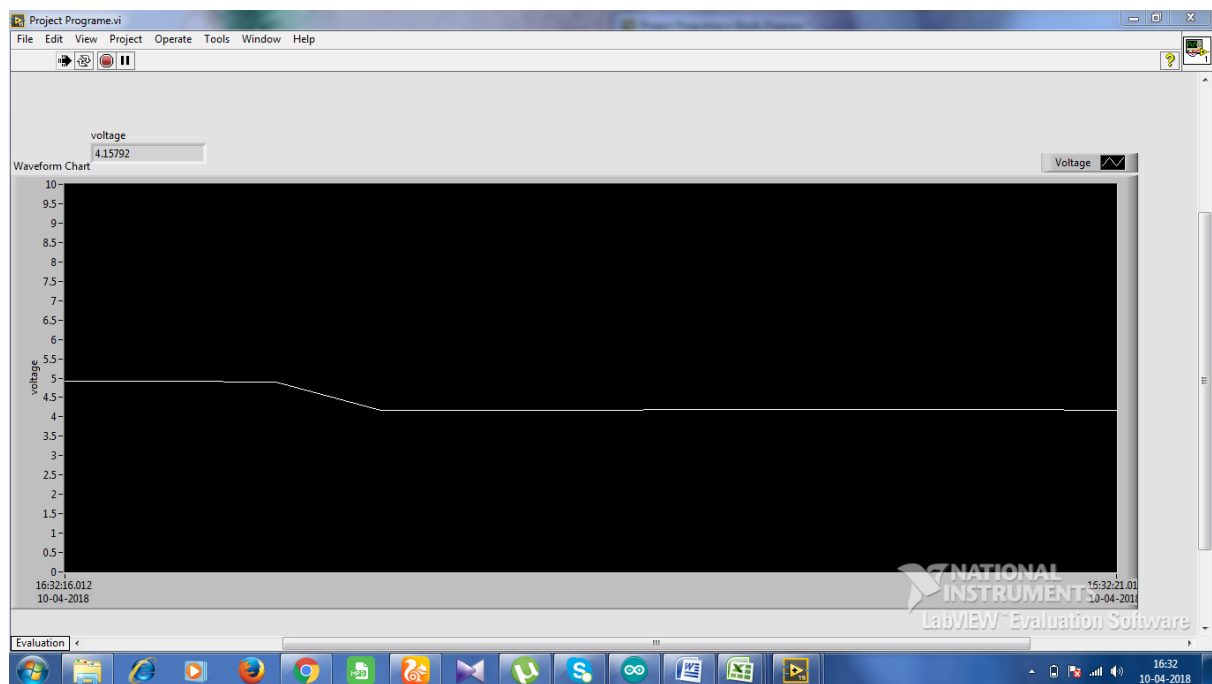
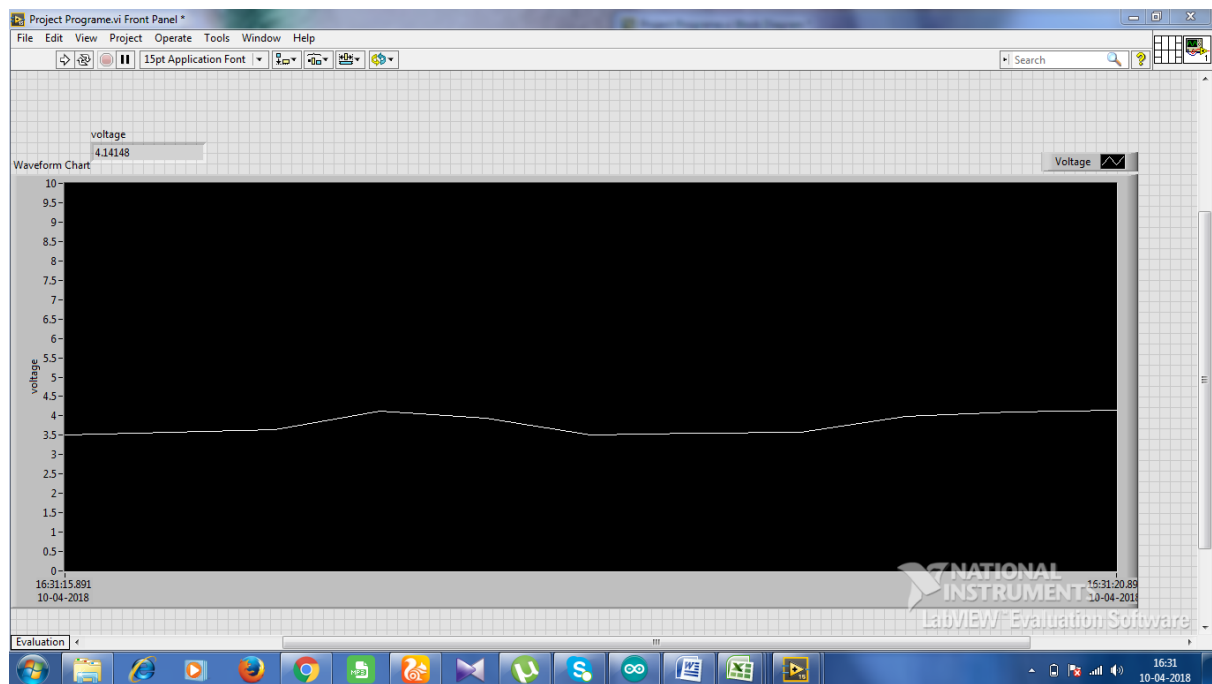
4.1 Electronics and LABVIEW design

A setup of the three linked gripper was manufactured as later in the figure. The servo motor was attached to each joint and the three pins of the servo motor were attached to the servo shield. The arduino was mounted at place such that it is free from interference from the links. The servo motors were programmed so that they coincide and get locked at a particular point together. By our calculation for each motor, the angle at which the lower links get locked at a particular position came out to be 60 degrees (for both upper and lower joint).

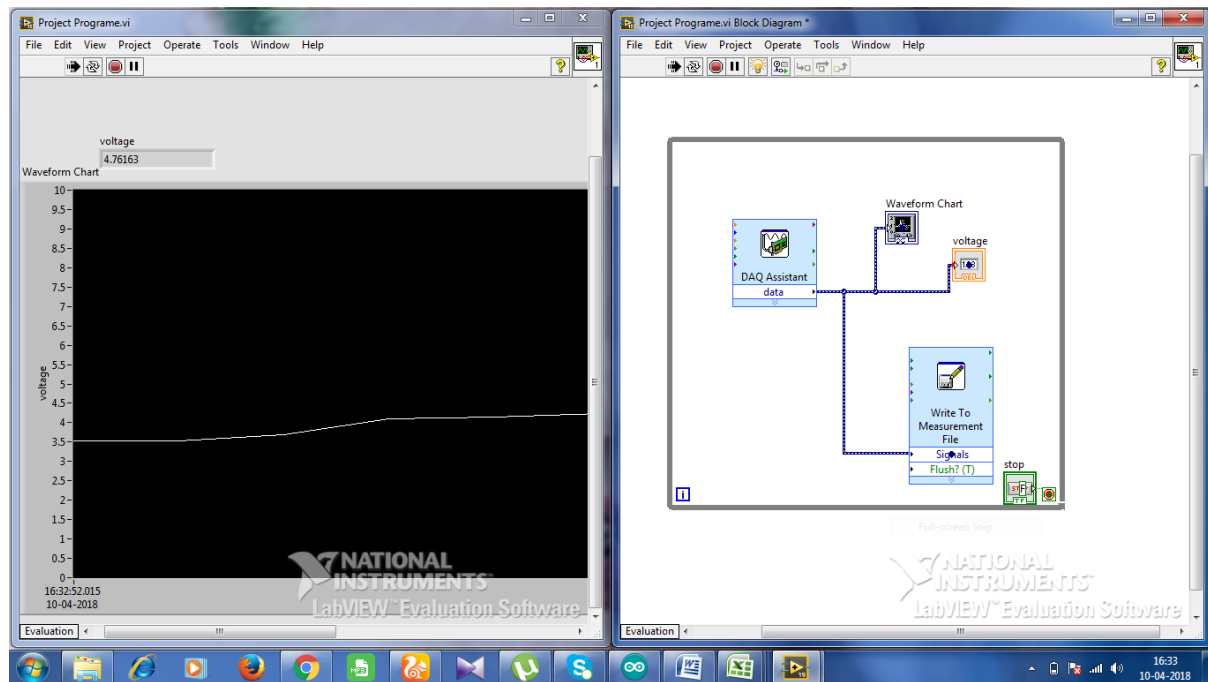
The arduino was given power and the links got locked at a fixed position. The setup which we used for our load analysis is as shown in the figure below. The load setup was made as light as possible so that it doesn't count with the existing load. The load setup was attached at the mouth of our gripper so that the lower links can grip it properly and symmetrically. Now, load was gradually applied to the gripper and for each load, voltage characteristics was studied and plotted which is shown later on. The hardware and software which we used for studying our voltage characteristics are ELVIS and LABVIEW. The voltage and ground wire out of one of the servo was attached to the ELVIS board and the ELVIS board was connected to our PC where we obtained our data in graphical plots in LABVIEW software. The block diagram of the circuit in LABVIEW is shown in the figure.



The panel through which we obtained our graphical readings is shown below:-



The combined view of our graphical panel and block diagram is shown below:-



4.2 Mechanical model

Until now we have fabricated a model of the gripper with each finger being given three links and two joints. We have attached the servo motors in their desired location and have checked the movement of each joint.

The constituent parts of the model are:

1. Cylindrical MS bar of a given length
2. Nine links made of aluminium with length 8.5 cm
3. Nut and bolts
4. A cylindrical rod of a fixed length having external threading upto some length and a similar nut . Both are made using hand threading tool
5. Alumunium bar
6. A tripod stand

4.3 Procedure followed in making the mechanical model of the gripper

1. Three links are kept as fixed and they are attached to the cylindrical MS bar using screws.
2. Holes are drilled in the links for the micro servo to fit in. First small holes are drilled using gun drilling machine and then the holes are enlarged using hand boring tool.
3. The links are attached accordingly and successively along with the servo motors forming a complete finger and we get three such fingers.
4. The cylindrical MS bar is made hollow by using lathe drilling operations
5. The aluminium bar is taken and holes are drilled on both ends.
6. A bolt is inserted through the hole of the cylindrical MS bar and attached at one end of the bar using a nut.
7. The long MS rod is taken and a small portion of it is made to undergo facing operations to decrease the diameter.

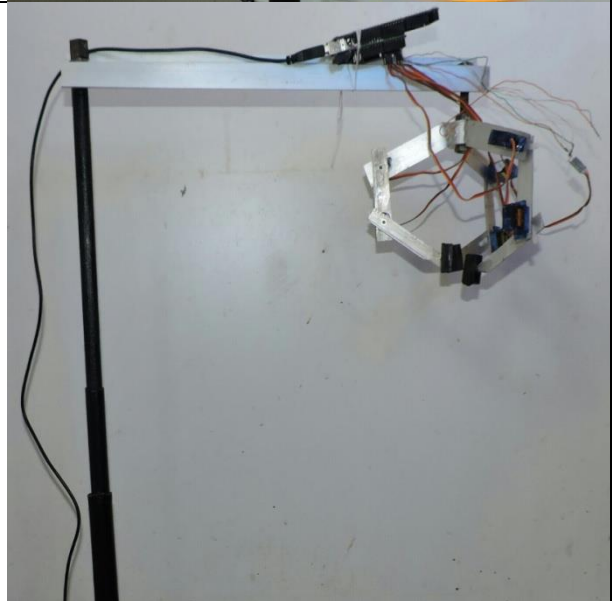
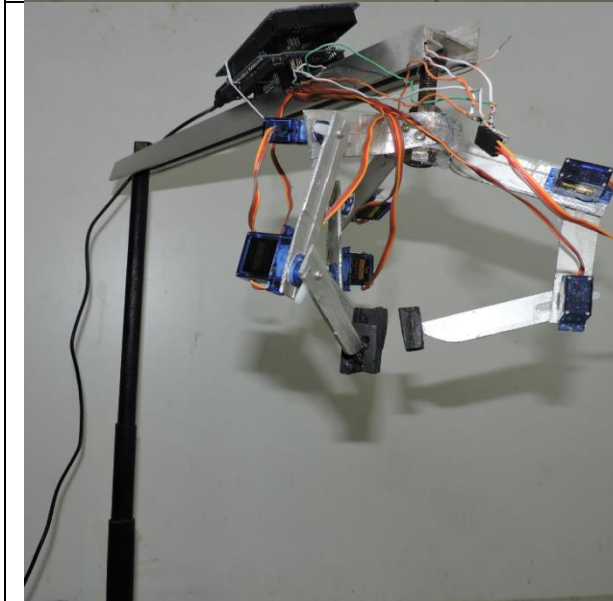
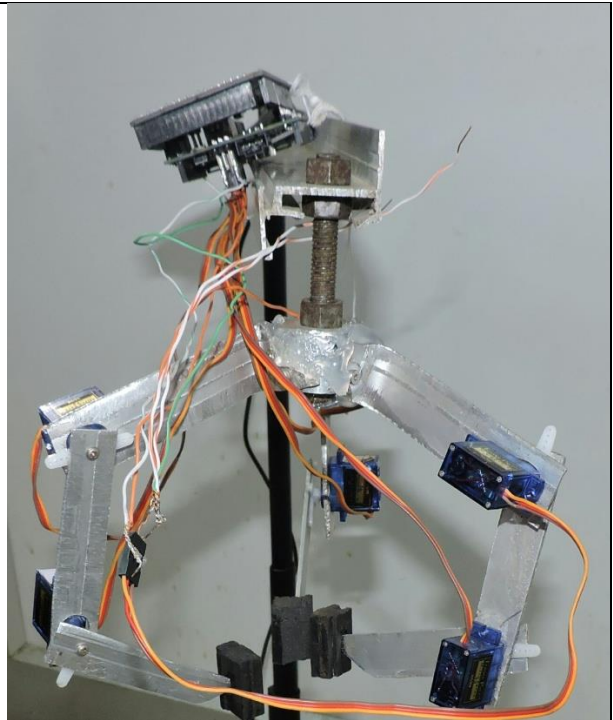
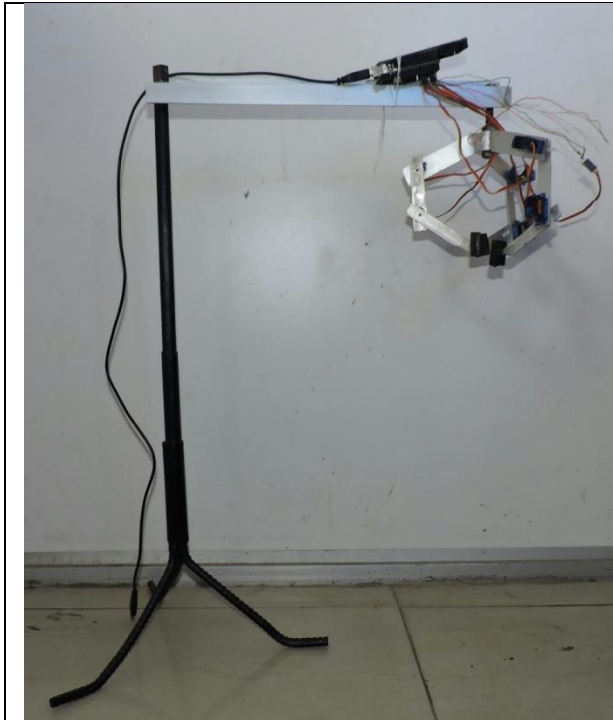
8. A hand threading tool is used to make external threads on that small portion of the MS rod.
9. In the same way a nut is made using a small MS bar which is drilled in lathe and internal threads are made by hand threading tools.
10. The rod and the frame are both attached at the threaded portion of the rod and fixed by using the handmade nut.
11. The rod is inserted into the tripod stand and the setup is ready.

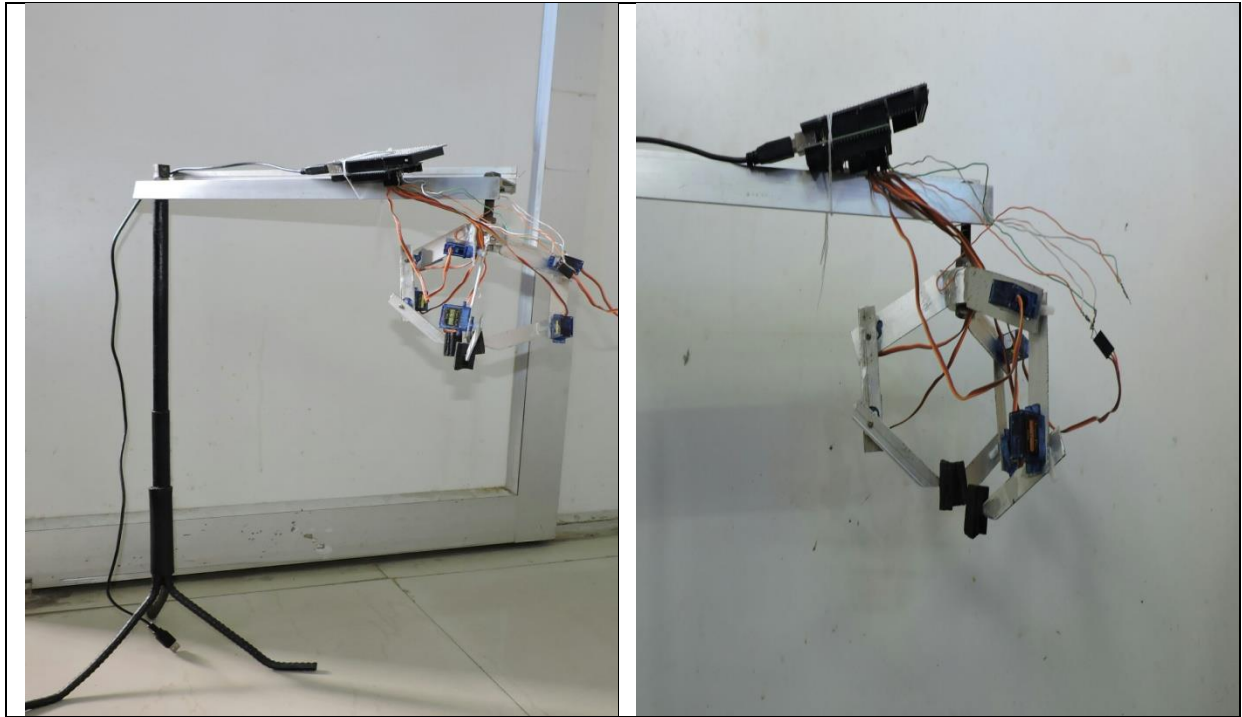


Hand made external threading on bolt



Hand made nut





VIEWS FROM DIFFERENT ANGLE OF THE FINAL ASSEMBLY

4.4 Programming

```
#include <Servo.h>
```

```
Servo myservo1;
```

```
Servo myservo2;
```

```
Servo myservo3;
```

```
Servo myservo4;
```

```
Servo myservo5;
```

```
Servo myservo6;
```

```
int pos =0;
```

```
void setup()
```

```
{
```

```
  myservo1.attach(1);
```

```
  myservo2.attach(2);
```

```
  myservo3.attach(3);
```

```
  myservo4.attach(4);
```

```
  myservo5.attach(5);
```

```
  myservo6.attach(6);
```

```
}
```

```
void loop() {
```

```
  myservo4.write(60);
```

```
  myservo5.write(60);
```

```
  myservo6.write(60);
```

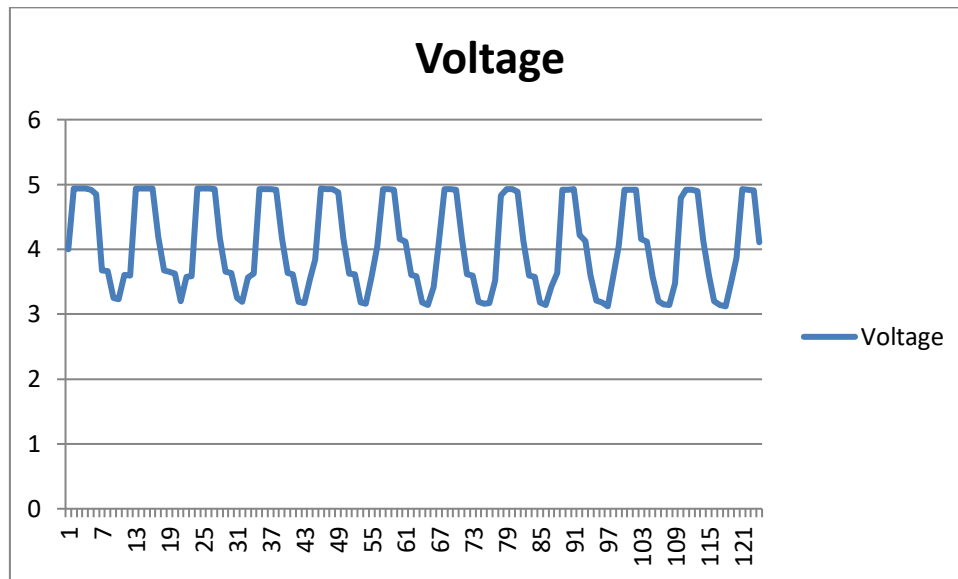
```
myservo1.write(60);  
myservo2.write(60);  
myservo3.write(60);  
}
```

CHAPTER 5:

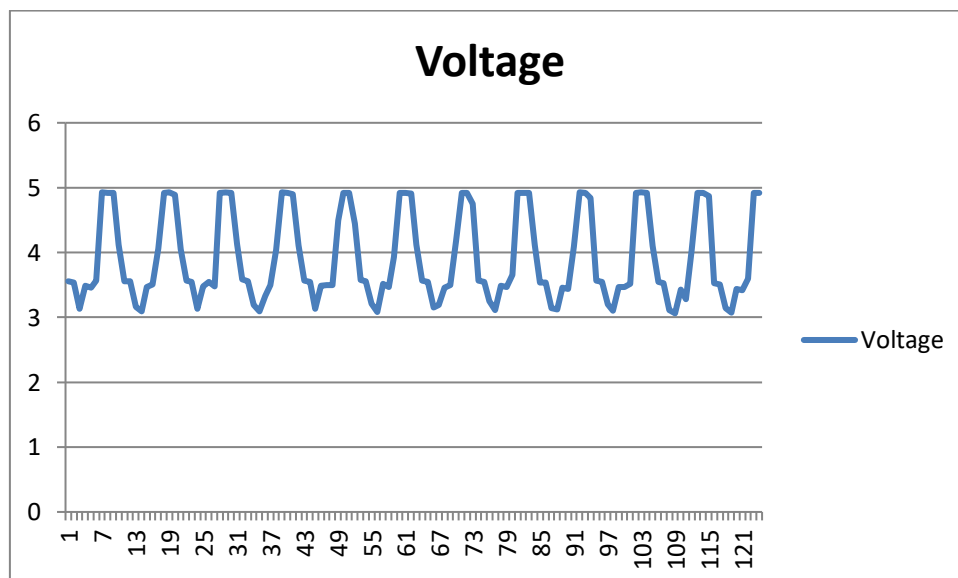
Experimental Results

Sampling frequency= 500Hz

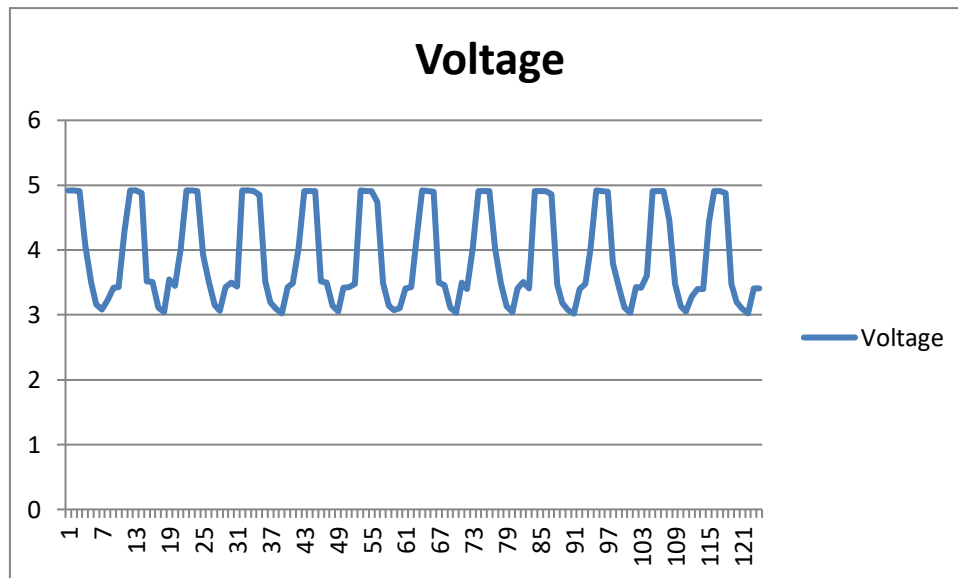
Voltage sampled for load applied= 0 grams



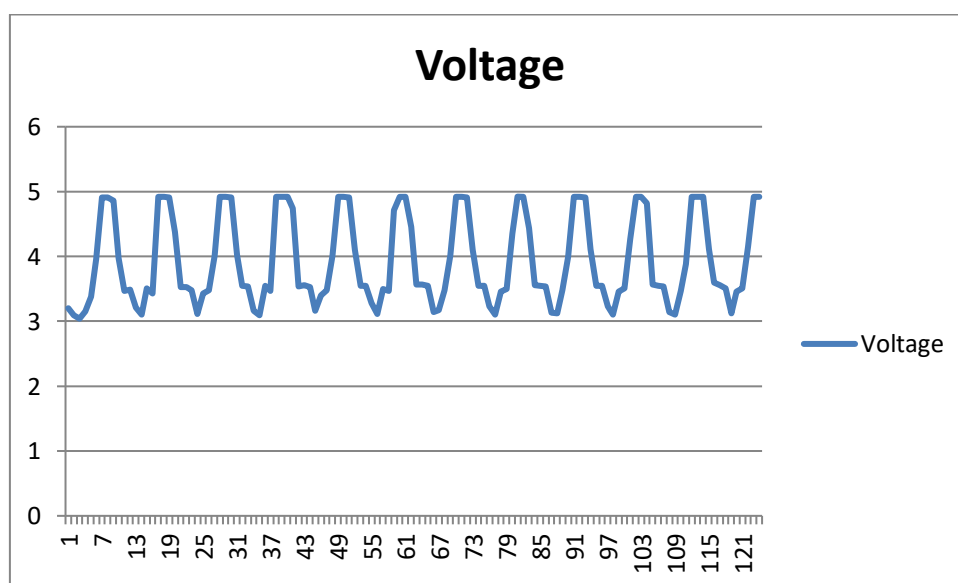
Voltage sampled for load applied= 5 grams



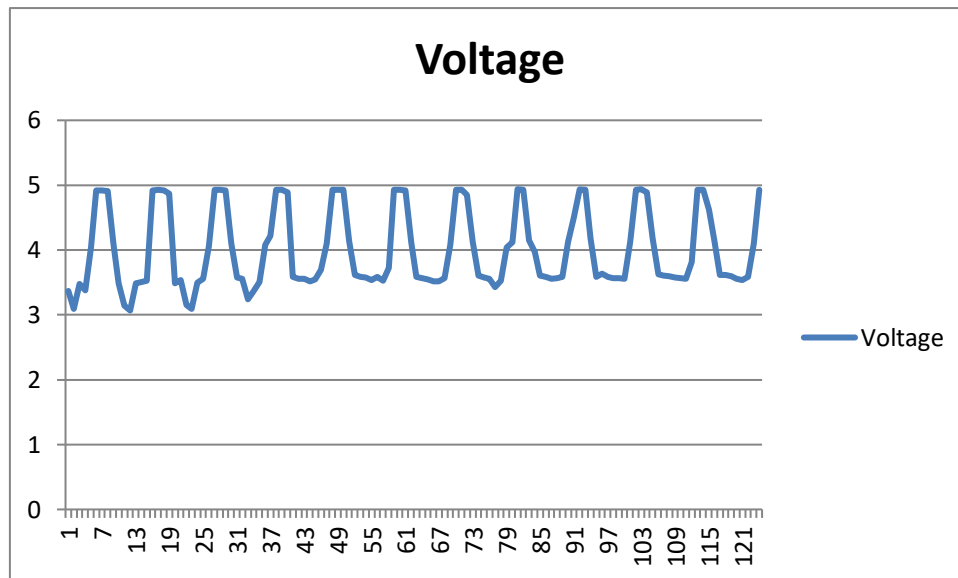
Voltage sampled for load applied= 10 grams



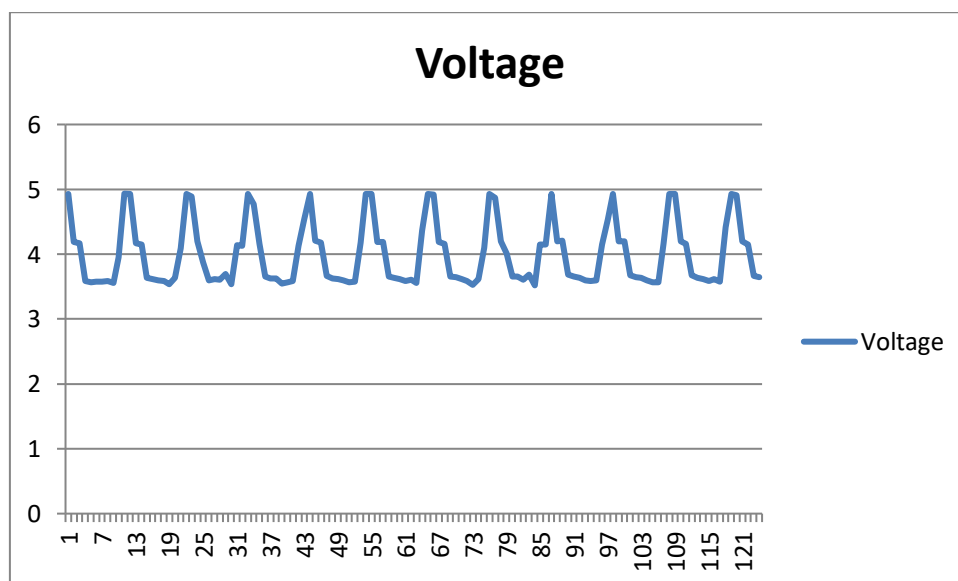
Voltage sampled for load applied= 15 grams



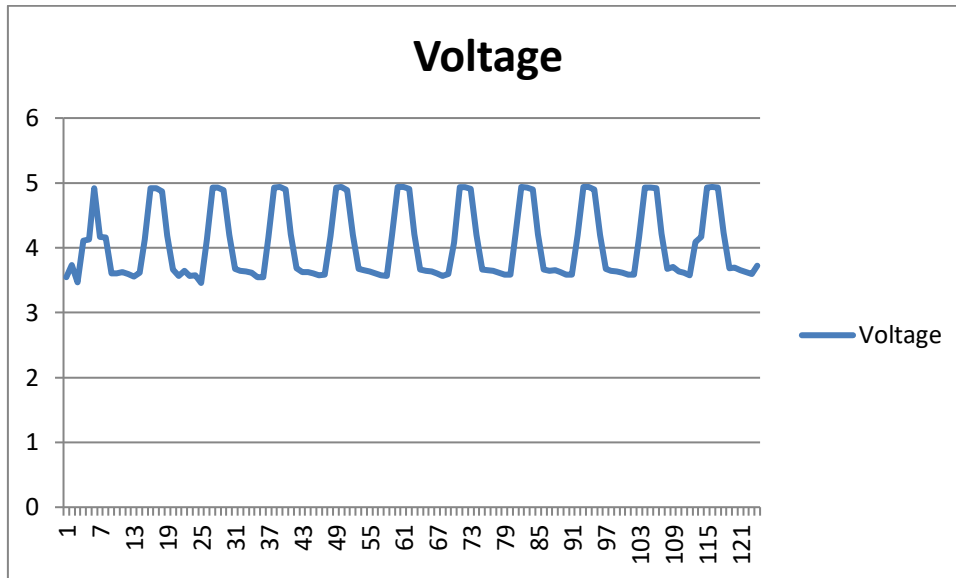
Voltage sampled for load applied= 20 grams



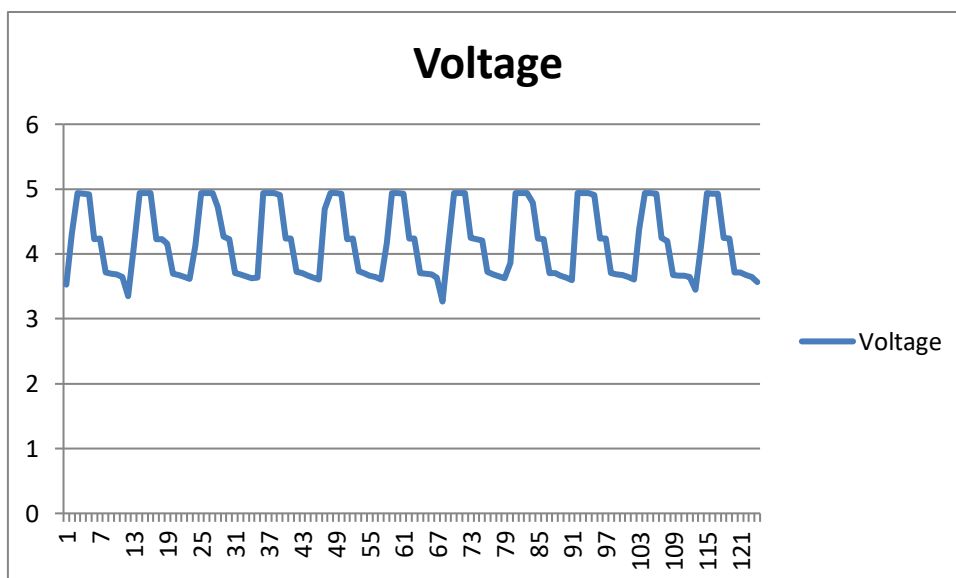
Voltage sampled for load applied= 25 grams



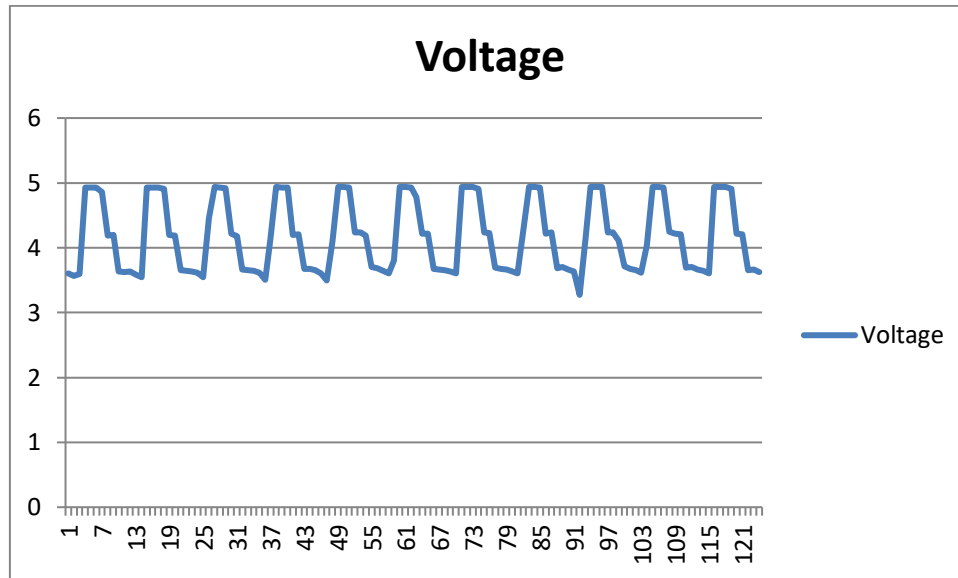
Voltage sampled for load applied= 30 grams



Voltage sampled for load applied= 35 grams

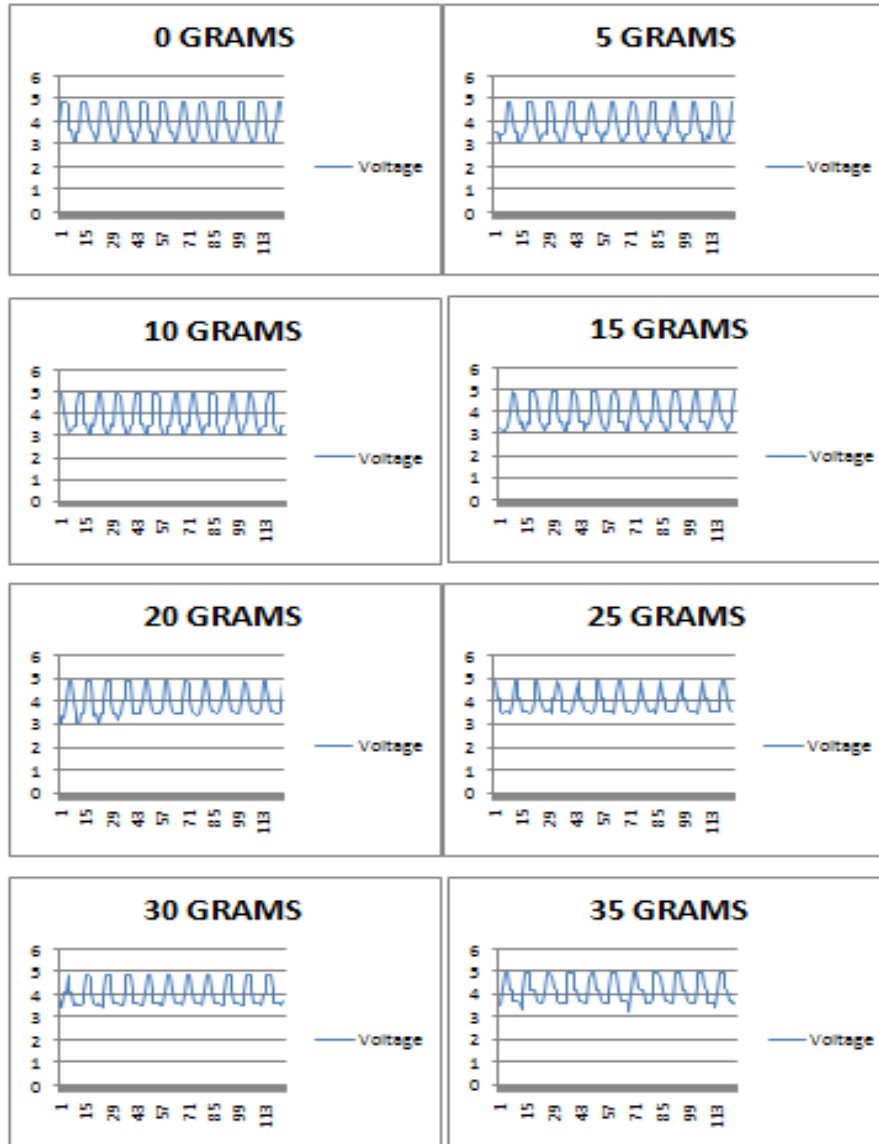


Voltage sampled for load applied= 40 grams



CHAPTER 6:

Theoretical explanation



From the above graphs, we can see that the graph is a continuous one having peaks and depressions at regular intervals. It increases to a fixed high value i.e. 4.92V and then starts decreasing until it reaches to some value which varies with the applied load. We can infer from the graphs that the minimum value of the curves increases considerably with increasing load.

Now the question is why we obtained such a pattern of graph and why the minima of the curve varied with the applied load. We know that the maximum input voltage of the servo motor which we have used in our project is 4.92V. The above voltage that has been stated is only under no- load condition. When the load was applied in our gripper system, all the motors experienced a certain amount of torque. Now, since the motor was unable to rotate

further, the voltage exceeded to some extent in amount of a few milli - volts and reached a peak value. After the peak value has been reached, the voltage was unable to increase any further and it started dissipating the voltage in the form of heat. The heat generated could be felt by touching the servo motors which was in experimental operation of the gripper system. The heat continued to dissipate until the motor had enough voltage to hold the object and apply sufficient torque. Hence the amount of heat dissipation varied with the applied load which can account for the reason behind different minimum value of the curve with different loads. It increased with the amount of load applied since more the load, smaller will be the heat dissipation as the motor need to have sufficient torque to grip the object which will be more for higher voltage.

After sufficient amount of heat has been generated, the voltage reached a value which was minimum for the motor to provide the minimum or sufficient torque after which the voltage started increasing again as it was trying to reach its maximum input voltage to provide the maximum torque. The process went on and we obtained the graph as shown above.

By analysing the graph carefully, we find that there are considerable amount of distortions that have taken place while obtaining the graphical plots. The distortions are irregular and they differ from each other in successive graphical plots. Hence we can deduce that they might have been generated from the noise and other disturbance from the surroundings which might have got added or got subtracted from the original voltage. Also it might have caused the delay that have resulted in generation of a constant voltage for a brief period of time.

Now, when we see the voltage versus time characteristics for a normal DC motor, we find that the voltage after going down from a maximum value increases to some fixed value after which it stabilizes. Now the question lies in the fact that since the servo motor is itself a DC motor, why we have a different curve for servo.

For explaining this phenomenon, we need to look deeper into the characteristics of the servo motor and see what actually is happening inside it. The servo motor consists of a DC motor along with a potentiometer and an electronic circuitry board which consists of a controller and amplifier. The servo motor operates by supplying PWM signals where the angle to be turned by the rotor depends on the width of the PWM signal. For a 90 degree rotation, the width of the pulse that we supply is generally 1.5ms. Pulses having width less than the above value will make the servo motor rotate in counter clockwise direction and width more than the same value will make the servo motor rotate in clockwise direction. When the rotor starts to rotate, the potentiometer attached to it using gearing arrangements starts to rotate and it generates another PWM pulse which is fed into the controller. Now the controller checks whether both the PWM pulse are same. If they are same, it stops the rotor in that fixed position which was specified by the PWM signal supplied by us. If the potentiometer pulse width is greater than our PWM signal width, then rotor moves to achieve the same width. Similarly, If the potentiometer pulse width is lesser than our PWM signal width, then rotor moves to achieve the same width. Hence what basically the controller does is that it sends an error signal by comparing the two PWM pulses. In the former case we call it a positive error signal and in the later case, we call it negative error signal.

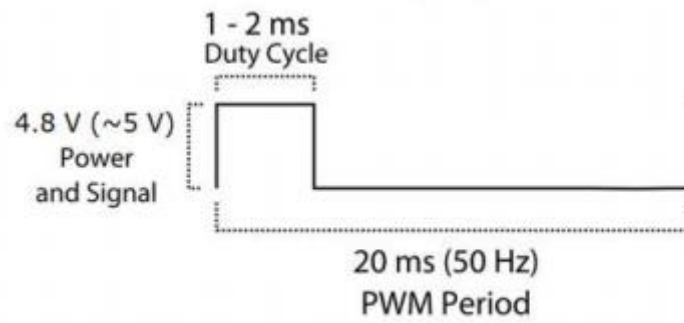


Fig: PWM pulse for micro servo SG90

Now, from the above theory let us try to analyse what is happening inside our servo motor. When our servo was in locked position, the controller used a fixed voltage to supply power to the DC motor and also there was a fixed amount of counter torque applied to the motor to keep it in position. When the load is applied, the rotor tends to rotate. In this case, the rotor tends to move in clockwise direction. Under normal operating condition the voltage used by the controller to supply to DC motor is 4.92V. As the rotor tends to move in clockwise direction, the controller would start increasing its voltage to provide more torque to counteract the applied load. Now, the maximum voltage that the controller can take is 4.93V and it can provide a maximum torque of $_V$. Hence, after reaching the maximum value, the controller cannot increase the voltage or torque anymore but the motor continues to draw power from the controller. Hence, heat starts to generate in the process which in turn causes the voltage to drop with time. As the value of the voltage decreases at some point of time the rotor rotates to some angle in the clockwise direction. This sudden clockwise rotation generates a change in PWM pulse width of the potentiometer and hence an error signal is generated. This signal is automatically fed to the controller which starts providing a counter torque in order for the rotor to reach the previous position. Hence, the voltage of the controller starts increasing again to provide the counter torque. It reaches the normal operating voltage i.e. 4.92 V and after that the process goes on.

Now that we have given some theory in defence of the above phenomenon, the thing which is left to be explained is the variation of the minimum voltage with applied load. We already know the reason behind the shape of the curve. When the load was applied and the voltage started decreasing, there was a point of time when the rotor started rotating when the error signal got generated by the controller. The difference in value of minimum voltage can be attributed to the change in time of generation of error signal. When the load applied is more, the lesser the time it will take for the rotor to start rotating and hence error signal will be generated sooner. After the error signal gets generated the voltage will start to increase as explained above and hence we get a variable minimum value for variable load.

CHAPTER 7

References

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